Low-Latency Rate-Distortion-Perception Trade-offs Through Randomized Distributed Function Computations

Onur Günlü[1,2], Maciej Skorski[3], H. Vincent Poor[4]

[1]Information Coding Division, Linköping University, Sweden.
[2]Lehrstuhl für Nachrichtentechnik, Technische Universität Dortmund, Germany.
[3]Faculty of Information Technology, Czech Technical University in Prague, Czech Republic.
[4]Department of Electrical and Computer Engineering, Princeton University, U.S.A.

Corresponding author: Onur Günlü, onur.gunlu@liu.se.

Abstract

Semantic communication systems focus on the transmission of meaning rather than the exact reconstruction of data, reshaping communication network design to achieve transformative efficiency in latency-sensitive and bandwidth-limited scenarios. Within this context, we investigate the rate-distortion-perception (RDP) problem for image compression, which plays a central role in producing perceptually realistic outputs under rate constraints. By employing the randomized distributed function computation (RDFC) framework, we derive an achievable non-asymptotic RDP region that captures the finite blocklength trade-offs among rate, distortion, and perceptual quality, aligning with the objectives of semantic communications. This region is further generalized to include either side information or a secrecy requirement, the latter ensuring strong secrecy against eavesdroppers through physical-layer security mechanisms and maintaining robustness in the presence of quantum-capable adversaries. The main contributions of this work are: (i) achievable bounds for non-asymptotic RDP regions subject to realism and distortion constraints; (ii) extensions to cases where side information is available at both the encoder and decoder; (iii) achievable, nonasymptotic RDP bounds with strong secrecy guarantees; (iv) characterization of the asymptotic secure RDP region under a perfect realism constraint; and (v) demonstrations of significant rate reductions and the effects of finite blocklengths, side information, and secrecy constraints. These findings offer concrete guidelines for the design of low-latency, secure, and high-fidelity image compression and generative modeling systems that generate realistic outputs, with relevance for, e.g., privacy-critical applications.

Keywords: randomized distributed function computation (RDFC), ultra-efficient semantic communication, realism for image compression, realism for generative modeling, rate-distortion-perception (RDP).

1 Introduction

Semantic communications represent a fundamental shift in how modern communication networks are designed and their objectives are defined. Rather than focusing on the exact reconstruction of transmitted signals, these systems emphasize conveying the underlying meaning of the data [1, 2]. This approach is especially advantageous in latency-sensitive and bandwidth-limited environments, such as augmented reality, autonomous driving, and immersive multimedia systems, where preserving only the semantically relevant information drastically reduces communication overhead while ensuring that the transmitted content remains functionally useful.

The semantic communication problem can be viewed as a special case of remote source coding problems [3, pp. 118], [4, pp. 78], [5], where the receiver's task is to compute a hidden function of the transmitter's observations. Building on this perspective, we recently proposed the randomized distributed function computation (RDFC) framework [6, 7], which explicitly incorporates the need for controlled randomization in many distributed computation scenarios. Such randomization, realized through preserving a target probability distribution, arises naturally in applications like neural compression with generative models [8, 9], federated learning with side information [10], and differentially private neural compression [7, 11, 12]. In contrast to adding artificial noise, RDFC employs coded strategies to synthesize the required randomness, ensuring that receiver outputs statistically follow a target distribution while maintaining utility. This design yields substantial improvements in communication efficiency, particularly for latency-sensitive and privacy-critical tasks. Furthermore, RDFC provides strong performance guarantees without relying on large amounts of common randomness shared between the transmitter and receiver, since the guarantees hold for each computation instance [7, 13, 14].

This work focuses on the rate–distortion–perception (RDP) problem, which arises in applications such as image compression [15, Section 17.4.2] and generative modeling [16]. As a particular instance of the RDFC framework, the RDP problem addresses the challenge of producing perceptually realistic image reconstructions while operating under rate constraints [17–21]. Unlike the classical rate–distortion formulation, RDP problem introduces an additional requirement on perceptual quality, ensuring that reconstructions not only minimize distortion but also correspond to human visual perception. Deep learning–based discriminators [22–24] are widely employed to generate images that cannot be distinguished from the originals. For analytical tractability, a common realism requirement is to enforce that the distribution of reconstructed images is close to that of the original source, as summarized in [25, 26]. This realism condition naturally fits within the RDFC framework, since the synthesized randomization step ensures that receiver outputs follow a distribution close to that of the transmitter's input. By casting the RDP problem within the RDFC framework, we employ coded RDFC techniques, providing strong function computation guarantees with limited or no shared randomness, to characterize non-asymptotic RDP limits in finite blocklength regimes.

In this work, we establish the achievable non-asymptotic trade-offs among rate, distortion, and perceptual quality in the RDP setting. The resulting finite blocklength characterizations serve both as theoretical reference points for evaluating practical coding schemes and as design guidelines, offering concrete insights for real-world scenarios that demand low-latency image compression. Moreover, we extend these bounds for the scenarios where both the encoder and decoder have access to data that are correlated with the input data, i.e., shared side information (SI). A possible use case of such a scenario is continuous image transmissions in which previous images contain (semantic) data that are correlated with the current image data.

Distributed function computations are often vulnerable to information leakage when communication takes place over public channels [27, 28]. To address this, we adapt our coding-theoretic constructions for the RDP problem by incorporating physical layer security (PLS) techniques, which minimize the information available to an eavesdropper. Thus, we analyze neural image compression settings where the channel output is accessible not only to the intended receiver but also to an adversary. Unlike conventional cryptographic approaches that rely on computational hardness, PLS methods offer security guarantees that remain valid even against quantum-capable attackers [29]. Furthermore, PLS plays an essential role in joint source–channel coding schemes for image compression [30], which naturally extend the RDP framework considered here. Within this context, we characterize both asymptotic and non-asymptotic secure RDP regions under strong secrecy constraints, where leakage becomes vanishingly small, in contrast to standard PLS approaches that only ensure weak secrecy by bounding normalized leakage.

Thus, incorporating coding-theoretic techniques from the RDFC framework into the RDP setting is an essential step toward connecting theoretical progress in distributed function computation, with strong computation and security guarantees, to practical schemes for neural image compression. By enabling perceptually optimized compression that operates with low latency while also ensuring secrecy, the proposed approach advances semantic-aware distributed function computation systems. In doing so, this work responds to the needs of modern industries that require communication solutions that are ultra-efficient, secure, and context-aware.

1.1 Main Contributions and Paper Organization

The main contributions of this work can be summarized as follows. We

- Derive an achievable non-asymptotic RDP region under realism and distortion constraints, providing inner bounds on required rates for finite blocklengths with strong perceptual quality guarantees;

- Generalize these results by incorporating an information-leakage constraint, establishing achievable bounds that provide strong secrecy guarantees while preserving perceptual quality and distortion performance;
- Characterize the asymptotic secure RDP region under the perfect realism constraint, demonstrating the relationship between near-perfect and perfect realism constraints;[1]
- Establish the non-asymptotic RDP with SI regions, where shared SI is correlated with the encoder input; and
- Analyze the resulting rate regions, which includes (i) illustrating significant communication load reductions over classical data compression methods; (ii) quantifying the effects of secrecy constraints on achievable rates; (iii) illustrating significant communication and common-randomness rate gains from available SI; and (iv) highlighting the relationships between non-asymptotic and asymptotic results.

1.2 Paper Organization

Section 2 introduces the point-to-point RDFC model considered in this work, designed to achieve high perceptual quality in neural image compression under distortion and communication constraints. This section also formalizes the definitions of non-asymptotic RDP regions, both with and without SI or secrecy. Section 3 presents the main contributions by deriving three achievable, finite-blocklength rate regions together with an asymptotic secure RDP region. In Section 4, we provide a comparative study, examining differences among the derived regions, their asymptotic limits, and classical data compression benchmarks. Section 5 outlines the proof sketches underlying the main results established in Section 3. Finally, Section 6 summarizes the findings and discusses their broader scientific and technological implications.

1.3 Notation

Random variables are denoted by uppercase letters X and their realizations by lowercase letters x. A random variable X follows a probability distribution P_X with support $\mathrm{supp}(P_X)$. Sets are represented using calligraphic letters $\mathcal{X}$ and their cardinality is written as $|\mathcal{X}|$. Sequences of length n are expressed as $X^n = (X_1, X_2, \dots, X_n)$. For an independent and identically distributed (i.i.d.) sequence, the joint distribution is given by $P_X^n \triangleq \prod_{i=1}^n P_X$.

The total variation (TV) distance between two distributions P_Y and P_X is defined as

$$\|P_Y - P_X\|_{\mathrm{TV}} \triangleq \frac{1}{2} \sum_{b \in \mathcal{B}} |P_Y(b) - P_X(b)|. \tag{1}$$

[1] Parts of these results appeared in the conference version of this work in [31], and here we corrected a minor issue in the distortion expressions.

$\mathcal{O}(\cdot)$ refers to the Big O notation. The information density associated with a distribution P_{XY} is

$$\imath(X,Y) = \log \frac{P_{XY}(x,y)}{P_X(x)P_Y(y)}. \tag{2}$$

Analogously, for a distribution of the form $P_{XYZ} = P_{XZ}P_Y$, the information density is defined as

$$\imath(XZ,Y) = \log \frac{P_{XZY}(x,z,y)}{P_{XZ}(x,z)P_Y(y)}. \tag{3}$$

The variance of a random variable is denoted by $\mathrm{Var}[\cdot]$, and $Q^{-1}(\cdot)$ represents the inverse Q-function, i.e., the tail distribution of a standard Gaussian. For any $k \in \mathbb{R}$, define $[k]^+ = \max k, 0$. All logarithms are taken to base 2. The notation $[a:b]$ represents the integer set $a, a+1, \ldots, b$, while $\cdot^c$ indicates the complementary event.

Finally, for any $\delta > 0$, a sequence x^n is called δ-letter typical with respect to P_X, denoted as $x^n \in T_\delta^n(P_X)$, if its empirical distribution $N(\cdot|x^n)/n$ satisfies

$$\left| \frac{N(a|x^n)}{n} - P_X(a) \right| \leq \delta P_X(a) \qquad \text{for all } a \in \mathcal{X}. \tag{4}$$

2 Problem Definition

We study a point-to-point RDFC problem aimed at achieving high perceptual quality while satisfying both communication-rate and reconstruction-distortion constraints, as illustrated in Fig. 1. The encoder observes an image $X^n \in \mathcal{X}^n$, where $\mathcal{X}$ is finite, and has access to common randomness $C \in [1 : 2^{nR_0}]$ shared with the decoder, which is uniformly distributed and independent of X^n. A practical source of such randomness is the use of physical unclonable functions (PUFs) [32], which allow the extraction and distribution of random sequences. The encoder maps (X^n, C) into an index $S \in [1 : 2^{nR}]$ through an encoding function $\mathsf{Enc}(\cdot)$. The index S is assumed to be received noiselessly at the decoder, which can be ensured by reliable transmissions using error-correcting codes. Based on (S, C), the decoder produces an image $Y^n = \mathsf{Dec}(S, C) \in \mathcal{X}^n$ such that:
(i) the resulting output distribution P_{Y^n}, with $y^n \sim P_{Y^n}$, closely matches the input image probability distribution Q_X^n;
(ii) the communication rate R is minimized for a given common randomness rate R_0; and
(iii) the distortion between X^n and Y^n is kept as small as possible.
We also consider a scenario where both the encoder and decoder have access to side information (SI) $Z^n \in \mathcal{Z}^n$, which is correlated with X^n such that $(X^n, Z^n) \sim Q_{XZ}^n$.

By fixing the blocklength $n \geq 1$, one can characterize the non-asymptotic limits of the RDP trade-off, which considers low-latency neural image compression. We therefore introduce three non-asymptotic regions for the considered RDFC problem: The first one without a secrecy constraint or SI, the second one with SI, and the third

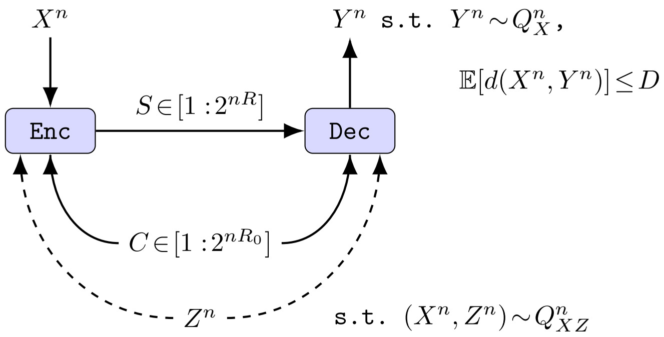

Fig. 1: An RDP model for deep learning–based image compression, where the realism constraint requires that the reconstructed image distribution at the receiver Y^n closely approximates the input image distribution Q_X^n, given $X^n \sim Q_X^n$. This condition guarantees high perceptual quality of the reconstruction. Moreover, the expected distortion between X^n and Y^n must be minimized to reduce the image quality loss caused by compression. To capture low-latency operations, we analyze the finite blocklength n cases and focus on minimizing the rate R for any fixed common randomness rate $R_0 \geq 0$. We also extend the model to scenarios in which SI Z^n, jointly distributed with X^n as $(X^n, Z^n) \sim Q_{XZ}^n$, is available to both the encoder and decoder.

one in which secrecy must be ensured against an eavesdropper observing the publicly transmitted index S. Throughout the following discussion, let $\epsilon_{\mathrm{r}}, \epsilon_{\mathrm{D}}, \epsilon_{\mathrm{sec}} > 0$ be arbitrary constants.

Definition 1 An RDP tuple (R, R_0, D) is $(\epsilon_{\mathrm{r}}, \epsilon_{\mathrm{D}}, n)$-*achievable* for Q_X if there exist an encoder and a decoder satisfying

$$\left\|P_{Y^n} - Q_X^n\right\|_{\mathrm{TV}} \leq \epsilon_{\mathrm{r}} \qquad \text{(realism)} \qquad (5)$$

$$\mathbb{E}\left[d(X^n, Y^n)\right] \leq D + \epsilon_{\mathrm{D}} \qquad \text{(distortion)} \qquad (6)$$

where $d(x^n, y^n) = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} d(x_i, y_i)$ is any per-letter distortion metric bounded from above by a value $d_{\max} > 0$.

The nonasymptotic RDP region $\mathcal{R}_{\mathrm{RDP}}$ is the closure of the set of all achievable tuples. $\Diamond$

We next define the nonasymptotic RDP with SI region $\mathcal{R}_{\mathrm{RDPSI}}$, where SI Z^n is correlated with X^n and is available to both the encoder and decoder.

Definition 2 Given shared SI Z^n, an RDP tuple (R, R_0, D) is $(\epsilon_{\mathrm{r}}, \epsilon_{\mathrm{D}}, n)$-*achievable* for Q_{XZ} if there exist an encoder and a decoder satisfying (5) and (6).

The nonasymptotic RDP with SI region $\mathcal{R}_{\mathrm{RDPSI}}$ is the closure of the set of all achievable tuples. $\Diamond$

We now impose a secrecy constraint that limits the information an eavesdropper, who observes the index S, can obtain about the reconstructed image Y^n. This aspect is especially important in applications such as generative artificial intelligence for producing artistic digital content, where the reconstructed image itself constitutes the valuable final output.

Definition 3 An RDP tuple (R, R_0, D) is $(\epsilon_{\rm r}, \epsilon_{\rm D}, \epsilon_{\rm sec}, n)$-*achievable* for Q_X under a strong secrecy constraint if there exist an encoder and a decoder satisfying (5), (6), and

$$||P_{SY^n} - P_S P_{Y^n}||_{\rm TV} \leq \epsilon_{\rm sec} \qquad\qquad \text{(strong secrecy)}. \tag{7}$$

The nonasymptotic secure RDP region $\mathcal{R}_{\rm SRDP}$ is the closure of the set of all achievable tuples. ♢

The constraint in (7) is a strong secrecy constraint, as it measures the unnormalized information leakage, unlike the classical weak secrecy constraint that considers a normalization by n, which is used, e.g., in [33].

We next provide non-asymptotic achievable RDP regions as inner bounds for $\mathcal{R}_{\rm RDP}$, $\mathcal{R}_{\rm RDPSI}$, and $\mathcal{R}_{\rm SRDP}$. Moreover, we characterize the asymptotic secure RDP region with a realism constraint that is more restrictive than (5).

3 Nonasymptotic Limits of RDP Trade-off

In line with [34–36], we introduce the channel dispersions corresponding to the channels $P_{U|X}$ and $P_{U|Y}$, respectively, as

$$V_{U|X} = \mathbb{E}_{P_{UX}}\big[\mathrm{Var}[\imath(U, X)|U]\big], \tag{8}$$
$$V_{U|Y} = \mathbb{E}_{P_{UY}}\big[\mathrm{Var}[\imath(U, Y)|U]\big]. \tag{9}$$

Here, $P_{U|X}$ and $P_{U|Y}$ denote test channels that connect the variables X and Y from the model in Fig. 1 to an auxiliary random variable U, which is used to represent the codebook construction. Furthermore, let

$$\mu_{xy} = \min_{(x,y)\in\mathrm{supp}(P_{XY})} P_{XY}(x, y). \tag{10}$$

We next characterize an $(\epsilon_{\rm r}, \epsilon_{\rm D}, n)$-achievable nonasymptotic RDP region $\mathcal{R}_{\rm RDP}$, whose proof outline is provided in Section 5.1.

Theorem 1 *An $(\epsilon_r, \epsilon_D, n)$-achievable nonasymptotic RDP region is the union over all distributions $P_{XUY} = Q_X P_{UY|X}$ of the rate tuples (R, R_0, D) such that*

$$R \geq \Big[I(U;X) + Q^{-1}\Big(\epsilon_r{+}\mathcal{O}\big(\frac{1}{\sqrt{n}}\big)\Big)\sqrt{\frac{V_{U|X}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+, \tag{11}$$

$$R + R_0 \geq \Big[I(U;Y) + Q^{-1}\Big(\epsilon_r{+}\mathcal{O}\big(\frac{1}{\sqrt{n}}\big)\Big)\sqrt{\frac{V_{U|Y}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+ \tag{12}$$

where $X - U - Y$ form a Markov chain, and

$$D \geq \mathbb{E}[d(X,Y)] - \delta_D - \epsilon'(\epsilon_r) d_{max} \tag{13}$$

with some function $\epsilon'(\epsilon_r) > 0$ such that $\epsilon'(\epsilon_r) \to 0$ if $\epsilon_r \to 0$ and

$$\epsilon_D = \delta_D(1 + D + \delta_D) + 2|\mathcal{X}|^2 e^{-2n\delta_D^2 \mu_{xy}^2} d_{max}. \tag{14}$$

It is sufficient to consider $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$.

The asymptotic counterparts of the bounds in (11)-(14) recover the RDP region in [25, Theorem 6] that extends [37, Theorems 1 and 5].

Remark 1 The asymptotic RDP region is obtained by letting $n \to \infty$ in Theorem 1, in which case a rate $R = I(U;X)$ becomes achievable provided that sufficient common randomness C is available, namely when $R_0 \geq I(U;Y) - I(U;X)$. Since $H(X)$ represents the minimum achievable rate for any lossless compression scheme, RDP methods can yield substantial improvements over classical compression methods, with gains quantified by the ratio $H(X)/I(U;X)$. For instance, [7] demonstrates that the use of common randomness in RDFC can reduce the rate by more than a factor of 214 in differential privacy applications. Consequently, depending on the underlying probability distribution Q_X and the particular distortion measure $d(\cdot,\cdot)$, analogous improvements can also be realized in the RDP setting.

Denote channel dispersions for test channels $P_{U|XZ}$ and $P_{U|YZ}$, respectively, as

$$V_{U|XZ} = \mathbb{E}_{P_{UXZ}}\big[\mathrm{Var}[\imath(U,XZ)|U]\big], \tag{15}$$

$$V_{U|YZ} = \mathbb{E}_{P_{UYZ}}\big[\mathrm{Var}[\imath(U,YZ)|U]\big]. \tag{16}$$

Let

$$\mu_{xy|z} = \min_{(x,y,z)\in \mathrm{supp}(P_{XY|Z})} P_{XY|Z}(x,y|z). \tag{17}$$

Next, we provide an $(\epsilon_\mathrm{r}, \epsilon_\mathrm{D}, n)$-achievable nonasymptotic RDP with SI region $\mathcal{R}_{\mathrm{RDPSI}}$. The proof outline of Theorem 2 is provided in Section 5.2.

Theorem 2 *An $(\epsilon_r, \epsilon_D, n)$-achievable nonasymptotic RDP with SI region is the union over all distributions $P_{XZUY} = Q_{XZ}P_{UY|XZ}$ of the rate tuples (R, R_0, D) such that (13) with*

$$\epsilon_D = \delta_D(1 + D + \delta_D) + 2|\mathcal{X}|^2|\mathcal{Z}| e^{-2n\delta_D^2 \mu_{xy|z}^2} d_{max} \tag{18}$$

and

$$R \geq \Big[I(U;X|Z) + Q^{-1}\Big(\epsilon_r + \mathcal{O}\big(\tfrac{1}{\sqrt{n}}\big)\Big)\sqrt{\frac{V_{U|XZ}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+, \tag{19}$$

$$R + R_0 \geq \Big[I(U;Y|Z) - H(Z|Y) + Q^{-1}\Big(\epsilon_r + \mathcal{O}\big(\tfrac{1}{\sqrt{n}}\big)\Big)\sqrt{\frac{V_{U|YZ}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+ \tag{20}$$

where $X - (U,Z) - Y$ form a Markov chain. It is sufficient to consider $|\mathcal{U}| \leq |\mathcal{X}|^2|\mathcal{Z}| + 1$.

We remark that by allowing $n \to \infty$, the bounds in (13) and (18)-(20) recover the asymptotic RDP with SI region in [38, Theorem 8].

Now, we provide an $(\epsilon_{\mathrm{r}}, \epsilon_{\mathrm{D}}, \epsilon_{\mathrm{sec}}, n)$-achievable nonasymptotic secure RDP region $\mathcal{R}_{\mathrm{SRDP}}$. The proof outline of Theorem 3 is provided in Section 5.3.

Theorem 3 *An $(\epsilon_r, \epsilon_D, \epsilon_{sec}, n)$-achievable nonasymptotic secure RDP region is the union over all distributions $P_{XUY} = Q_X P_{UY|X}$ of the rate tuples (R, R_0, D) such that, for any $\theta \in [0,1]$, (13), (14), and*

$$R \geq \Big[I(U;X) + Q^{-1}\Big(\theta\big(\epsilon_r + \mathcal{O}(\frac{1}{\sqrt{n}})\big)\Big)\sqrt{\frac{V_{U|X}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+, \tag{21}$$

$$R_0 \geq \Big[I(U;Y) + Q^{-1}\Big((1-\theta)\big(\epsilon_{sec} + \mathcal{O}(\frac{1}{\sqrt{n}})\big)\Big)\sqrt{\frac{V_{U|Y}}{n}} + \mathcal{O}\Big(\frac{\log n}{n}\Big)\Big]^+ \tag{22}$$

where $X - U - Y$ form a Markov chain. It is sufficient to consider $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$.

By letting $n \to \infty$, the asymptotic forms of the bounds in (13), (14), (21), and (22) coincide with the asymptotic secure RDP region derived in [39]. This recovery follows once the factors θ and $(1-\theta)$ in (21) and (22) are removed.

The constraint in (5) is often referred to as a *near-perfect* (or *strong*) realism constraint, because ϵ_{r} can be made arbitrarily small as $n \to \infty$, although it never reaches exactly zero. In contrast, the *perfect realism* constraint enforces

$$\|P_{Y^n} - Q_X^n\|_{\mathrm{TV}} = 0 \qquad \text{(perfect realism)} \tag{23}$$

for which the following asymptotic characterization holds.

Theorem 4 *The asymptotic secure RDP region with perfect realism constraint is the union over all distributions $P_{XUY} = Q_X P_{UY|X}$ of the rate tuples (R, R_0, D) such that*

$$R \geq I(U;X), \tag{24}$$

$$R_0 \geq I(U;Y), \tag{25}$$

$$D \geq \mathbb{E}[d(X,Y)] \tag{26}$$

where $X - U - Y$ form a Markov chain. It is sufficient to consider $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$.

Proof Outline: The argument follows by first taking the asymptotic limits of the bounds in (13), (14), (21), and (22) by letting $n \to \infty$, which yields the asymptotic secure RDP region given in [39]. This result is then combined with [26, Theorem 1], which establishes that the rate tuples (R, R_0, D) are achievable under near-perfect realism if and only if they are achievable under perfect realism. Therefore, merging the findings of [39] with [26, Theorem 1] establishes Theorem 4. □

Remark 2 The optimal code constructions for the asymptotic secure RDP regions under near-perfect and perfect realism are not necessarily the same although they have the same bounds.

4 Rate Region Comparisons and Analysis

4.1 Effects of SI

The presence of SI Z^n at both terminals can enlarge the achievable RDP region in two complementary ways. First, SI can reduce the communication rate R, i.e., compare (11) with (19). A similar effect carries over to the leading term of the sum-rate bound in (20). Second, the portion of Z^n that Y^n cannot infer acts as an additional common randomness, which can further reduce the common-randomness rate, i.e., compare (12) with (20). This dual role of SI, discussed also in [38], is what Theorems 1 and 2 capture in our non-asymptotic settings. In the special case where Z^n is independent of (X^n, C), SI does not reduce the communication rate but is equivalent to an independent common randomness source of rate $H(Z)$. Then, the achievable RDP with SI region in Theorem 2 reduces to the achievable RDP region in Theorem 1 with the difference that the required common randomness rate can be reduced by $H(Z)$; see also [38, pp. 10].

4.2 Effects of Secrecy Constraint

Compared with the non-secure RDP region in Theorem 1, the secure RDP region in Theorem 3 differs both in structure and in the resulting bounds. Without secrecy constraint, the description involves a single sum-rate constraint, which yields a larger achievable region. With secrecy constraint, by contrast, the rates are bounded separately to meet realism and secrecy simultaneously. Moreover, the finite-blocklength penalties grow, because the realism ϵ_{r} and secrecy ϵ_{sec} parameters are scaled by θ and $(1-\theta)$. Consequently, the additive terms are larger than in the non-secure setting. Taken together, these features underscore the cost of strong secrecy: It generally requires higher communication and common-randomness rates, with the penalty amplified at short blocklengths. This trade-off is critical in applications, e.g., that must balance security, latency, distortion, and realism when selecting between secure and non-secure neural compression methods.

4.3 Effects of Nonasymptotics

As $n \to \infty$, the finite blocklength penalties vanish, and the asymptotic rate regions are recovered. In non-asymptotic regimes, however, the achievable bounds contain terms of the form $Q^{-1}(\epsilon)\sqrt{V/n}$, capturing finite-length penalties arising from statistical fluctuations inherent at finite blocklengths. These additional terms share the same functional form as in finite blocklength channel and source coding [34, 40]. Comparing the nonasymptotic achievable RDP regions with their asymptotic counterparts, thus, clarifies how finite blocklength constraints affect the RDP trade-offs: At finite n, the constraints on rate and distortion are stricter than in the asymptotic case. The insights from finite blocklength analysis are crucial for establishing reference performance baselines for, e.g., neural image compression systems that must operate efficiently under stringent latency, realism, and distortion requirements.

5 Proof Outlines of Theorems 1-3

5.1 Proof Outline of Theorem 1

Proof Outline: The achievability proof is built on the nonasymptotic binning methods proposed and used, for instance, in [36, 41, 42], which develop finite-length code constructions as in [34, 40, 43] for the output statistics of random binning (OSRB) method [44, 45]. Below, we detail the specific adaptations and technical refinements used here, emphasizing where our approach departs from standard treatments in the literature.

Fix a distribution $P_{XUY} = Q_X P_{UY|X}$ such that

$$\mathbb{E}[d(X,Y)] \leq D + \delta_D \tag{27}$$

where $\delta_D \geq 0$ satisfies (14). We remark that the difference between the expected distortion $\mathbb{E}[d(X^n, Y^n)]$ under P_{XY}^n and under the synthesized joint probability distribution can be bounded by $\epsilon'(\epsilon_{\mathrm{r}}) d_{\max}$, where $\epsilon'(\epsilon_{\mathrm{r}}) > 0$ is independent of n and $\epsilon'(\epsilon_{\mathrm{r}}) \to 0$ if $\epsilon_{\mathrm{r}} \to 0$. This bound follows from similar arguments to [38, Section IV-G and Proposition 53][2]. Thus, in the following analysis, we consider i.i.d. sequences by accounting for this difference. Define the error event that the sequences (X^n, Y^n) are not δ_{D}-letter typical as

$$\mathcal{E} = \{(X^n, Y^n) \notin T^n_{\delta_{\mathrm{D}}}(P_{XY})\}. \tag{28}$$

By applying similar steps as in [46], we obtain (14), given (6), as we have

$$\begin{aligned}
\mathbb{E}[d(X^n, Y^n)] &= \Pr[\mathcal{E}]\ \mathbb{E}[d(X^n, Y^n)|\mathcal{E}] + \Pr[\mathcal{E}^c]\ \mathbb{E}[d(X^n, Y^n)|\mathcal{E}^c] \\
&\overset{(a)}{\leq} \Pr[\mathcal{E}]\ d_{\max} + \Pr[\mathcal{E}^c]\ (1+\delta_{\mathrm{D}})\ \mathbb{E}[d(X,Y))] \\
&\overset{(b)}{\leq} 2|\mathcal{X}|^2 e^{-2n\delta_{\mathrm{D}}^2 \mu_{xy}^2}\ d_{\max} + (1+\delta_{\mathrm{D}})\ (D+\delta_{\mathrm{D}})
\end{aligned} \tag{29}$$

where (a) follows from the typical average lemma [47, pp. 26] and because the distortion metric is per-letter with bound $d_{\max}$, and (b) follows from the bound on the probability of the error event $\mathcal{E}$ given in [48, Eq. (6.34)], applied since a per-letter estimator is used, and by (27).

We next prove that there exist non-asymptotic random binning schemes simultaneously satisfying the realism and distortion constraints. Generate an auxiliary random variable sequence U^n in an i.i.d. manner such that we have a joint probability distribution $P_{XU} = Q_X P_{U|X}$. Following the structure of the OSRB method, we first analyze a source coding problem (Protocol A). In Protocol A, the encoder maps U^n independently and uniformly to three random bin indices $S \in [1\!:\!2^{nR}]$, $F \in [1\!:\!2^{n\widetilde{R}}]$, and $C \in [1\!:\!2^{nR_0}]$. In this protocol, the index F represents the public choice of encoder-decoder pairs. Using a mismatch stochastic likelihood coder as the decoder, as in [41, pp. 3] and [36, Eq. (12)], we bound the expected error probability averaged over the random binning ensemble.

Now, the rate constraints are imposed to ensure that the encoder-decoder pair aims to satisfy the following, with the penalties for finite blocklengths: i) (C, F) are almost independent of X^n; ii) (C, F, S) almost recover U^n; and iii) F is almost independent of Y^n. To impose constraints ensuring near independence, we apply [41, Theorem 1]. Similarly, to impose reliable sequence reconstruction constraints, we apply [41, Theorem 2]. These steps yield rate

[2] [38, Proposition 53] introduces two additional error terms, beyond the realism parameter ϵ_{r}, which must also vanish in order to guarantee that $\epsilon'(\epsilon_{\mathrm{r}}) \to 0$. While this vanishing behavior constitutes a necessary condition, any asymptotically optimal code design will, by construction, satisfy it. For this reason, and in the interest of clarity, we provide only a sufficient condition for ensuring $\epsilon'(\epsilon_{\mathrm{r}}) \to 0$, without explicitly specifying the exact dependence on all parameters. See [38, Proposition 53] for the exact error terms.

constraints on $(\widetilde{R}, R, R_0)$, derived by applying Berry-Esseen Theorem such that total variation distances between the target and observed probability distributions are bounded by a fixed value. This analysis corresponds to Protocol B, a channel coding problem dual to our problem with extra randomness F. Furthermore, the proof of the realism constraint (5) follows by applying the soft covering lemma [13, Lemma IV.1] as in the achievability proof of [26, Theorem 2].

To eliminate the extra randomness F such that $\widetilde{R}$ is also eliminated from the rate constraints, we show that a fixed realization $F = f$ can be agreed upon publicly by the encoder and decoder, following by applying arguments similar to in [36, 45]. Finally, by selecting the free parameters similar to the choices in [36, Eq. (36)], we obtain (11) and (12).

The cardinality bound on the auxiliary random variable U follows by using the support lemma [3, Lemma 15.4]. We preserve the distribution P_{XY} by using $(|\mathcal{X}|^2 - 1)$ continuous real-valued functions, as we have $\mathcal{Y} = \mathcal{X}$. We must preserve two more expressions that are the lower bounds in (11) and (12). Thus, it suffices to have $|\mathcal{U}| \leq |\mathcal{X}|^2 + 1$. □

5.2 Proof Outline of Theorem 2

Proof Outline: The achievability proof follows by using a coding method similar to the code construction designed in Section 5.1, where, this time, we use a different codebook for each realization $z^n \in \mathcal{Z}^n$. Using a different codebook per realization is similarly applied to an asymptotic RDP with SI case in [38, Section IV-B], which leverages a strong coordination with SI result in [13, Corollary VII.5].

We then impose rate constraints to asymptotically satisfy the following: i) (C, F) are almost independent of (X^n, Z^n); ii) (C, F, S, Z^n) almost recover U^n; and iii) using the soft-covering lemma with SI [13, Corollary VII.5] and recognizing that the term $(I(U;Y|Z) - H(Z|Y))$ in (20) is equal to $(I(U,Z;Y) - H(Z))$ where the latter is the form used in [13, Corollary VII.5], we satisfy the realism constraint (5); see also [38, pp. 10-11]. Moreover, the elimination of the extra randomness F and the proof of the cardinality bound follow similarly to the steps given in Section 5.1. □

5.3 Proof Outline of Theorem 3

Proof Outline: The achievability proof follows similarly to the achievability proof of Theorem 1 given in Section 5.1. However, we here need to demonstrate that there exist non-asymptotic random binning schemes that simultaneously satisfy

$$\left\|P_{Y^n} - Q_X^n\right\|_{\mathrm{TV}} \leq \theta\epsilon_{\mathrm{r}}, \tag{30}$$

$$||P_{SY^n} - P_S P_{Y^n}||_{\mathrm{TV}} \leq (1-\theta)\epsilon_{\mathrm{sec}} \tag{31}$$

for any $\theta \in [0, 1]$; see also [41, Theorem 4]. We apply a similar random code construction as in Section 5.1, but rather than imposing that F is almost independent of Y^n we impose that (S, F) are almost independent of Y^n. Applying steps similar to those in Section 5.1, we obtain (21) and (22). Moreover, the cardinality bound follows similarly to Section 5.1 by preserving P_{XY} and the lower bounds in (21) and (22). □

6 Conclusions

Within the RDFC framework, this paper has extended classical rate–distortion trade-offs by incorporating perceptual quality and strong secrecy constraints. In particular, we established achievable non-asymptotic RDP regions and examined their asymptotic counterparts, together providing foundations for low-latency, high-fidelity, and secure neural image compression with realistic outputs. Moreover, we illustrated that shared SI can enlarge the achievable regions by simultaneously reducing the required communication and common randomness rates. Although the i.i.d. assumption for input sequences is idealized, it offers insightful theoretical benchmarks; while extensions to non-i.i.d. models via, e.g., information-spectrum methods [49] are possible, they typically lead to cumbersome expressions, so we focused on i.i.d. models to balance rigor and practical interpretability.

From a technological standpoint, the derived bounds and trade-offs inform the design of secure and ultra-efficient neural image compression systems for realism- and privacy-sensitive applications. By ensuring security against quantum-capable adversaries, the results align with the transition to quantum-safe communications. Natural directions for future work include incorporating noisy transmissions, which would deepen the theoretical understanding of joint RDP–channel coding methods and further enhance the applicability of RDP methods in emerging deep learning–based systems.

Declarations

Availability of data and materials

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Competing interests

The authors declare that they have no competing interests.

Funding

This work of OG was supported by the ZENITH Research and Leadership Career Development Fund under Grant ID23.01, EU COST Action 6G-PHYSEC, and the Swedish Foundation for Strategic Research (SSF) under Grant ID24-0087. The work of HVP was supported in part by the U.S National Science Foundation under Grant ECCS-2335876.

Authors' contributions

All authors participated in the conceptualization, methodology, mathematical analysis, and project management. OG participated in writing of the original draft, and MS and HVP participated in the reviewing and editing of the original draft.

Authors' information

OG received the B.Sc. degree (Highest Distinction) in Electrical and Electronics Engineering from Bilkent University, Turkey in 2011; M.Sc. (Highest Distinction) and Dr.-Ing. (Ph.D. equivalent) degrees in Communications Engineering both from the

Technical University of Munich (TUM), Germany in 2013 and 2018, respectively. He was a Working Student in the Communication Systems division of Intel Mobile Communications (IMC), now Apple Inc., in Munich, Germany during November 2012 - March 2013. Onur worked as a Research and Teaching Assistant at TUM Chair of Communications Engineering (LNT) between February 2014 - May 2019. As a Visiting Researcher, among more than twenty Research Stays at Top Universities and Companies, he was at TU Eindhoven, Netherlands during February 2018 - March 2018. Onur was a Visiting Research Group Leader at Georgia Institute of Technology, Atlanta, USA during February 2022 - March 2022. He was also a Visiting Professor at TU Dresden, Germany during February 2023 - March 2023. Following Research Associate and Group Leader positions at TUM, TU Berlin, and the University of Siegen, he joined Linköping University in October 2022 as an ELLIIT Assistant Professor and obtained tenure as an Associate Professor leading the Information Theory and Security Laboratory (ITSL) in August 2024. He obtained a Swedish Docent (Habilitation) of Information Theory title in December 2023 and became an IEEE Senior Member in July 2024. Since September 2025, Onur has been a Full Professor and Institute Head at the Lehrstuhl für Nachrichtentechnik at TU Dortmund and a Guest Professor at Linköping University.

He has received the 2025 IEEE Information Theory Society - Joy Thomas Tutorial Paper Award, the 2023 ZENITH Research and Career Development Award, 2021 IEEE Transactions on Communications - Exemplary Reviewer Award, and the prestigious VDE Information Technology Society (ITG) 2021 Johann-Philipp-Reis Award. His research interests include distributed function computation, information-theoretic privacy and security, coding theory, integrated sensing and communication, and private learning. Among his publications is the book *Key Agreement with Physical Unclonable Functions and Biometric Identifiers* (Dr. Hut Verlag, 2019). He serves as Associate Editor for IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, IEEE TRANSACTIONS ON COMMUNICATIONS, and ENTROPY, recently was an Associate Editor of EURASIP JOURNAL ON WIRELESS COMMUNICATIONS AND NETWORKING and Guest Editor of IEEE JOURNAL ON SELECTED AREAS IN INFORMATION THEORY. He also serves as a Board Member and Secretary of the IEEE Sweden VT/COM/IT Joint Chapter, and as a Working Group Leader for EU COST Action 6G Physical Layer Security (6G-PHYSEC).

MS was awarded a Ph.D. in 2023 from the University of Warsaw. He has over 15 years of research experience in academic and industrial roles, with interests spanning cryptography, information theory, algorithms, and machine learning. He currently works as a postdoctoral researcher at the Czech Technical University in Prague.

HVP received the Ph.D. degree in EECS from Princeton University in 1977. From 1977 until 1990, he was on the faculty of the University of Illinois at Urbana-Champaign. Since 1990, he has been on the faculty at Princeton, where he is currently the Michael Henry Strater University Professor. During 2006 to 2016, he served as the Dean of Princeton's School of Engineering and Applied Science, and he has also held visiting appointments at several other universities, including most recently at Berkeley and Caltech. His research interests are in the areas of information theory, machine learning and network science, and their applications in wireless networks, energy systems, and related fields. Among his publications in these areas is the book *Machine Learning and Wireless Communications* (Cambridge University Press, 2022). Dr. Poor is a member of the National Academy of Engineering and the National Academy of Sciences and is a foreign member of the Royal Society and other national and international academies. He received the IEEE Alexander Graham Bell Medal in 2017.

References

[1] D. Gündüz, et al., Beyond transmitting bits: Context, semantics, and task-oriented communications. IEEE J. Sel. Areas Commun. (JSAC) (2022)

[2] D. Gündüz, et al., Timely and massive communication in 6G: Pragmatics, learning, and inference. IEEE BITS Inf. Theory Mag. (2023)

[3] I. Csiszár, J. Körner, *Information Theory: Coding Theorems for Discrete Memoryless Systems* (Cambridge University Press, 2011)

[4] T. Berger, Rate-distortion theory. Wiley Encyclopedia of Telecommunications (2003)

[5] R. Dobrushin, B. Tsybakov, Information transmission with additional noise. IRE Trans. Inf. Theory (T-IT) **8**(5), 293–304 (1962)

[6] O. Günlü, Randomized distributed function computation (RDFC): Ultra-efficient semantic communication applications to privacy. EURASIP J. Inf. Security (JIS) (2025). Submitted

[7] O. Günlü, *Randomized Distributed Function Computation with Semantic Communications: Applications to Privacy*, in *Proc. IEEE Int. Workshop Inf. Forensics Security (WIFS)* (2024), pp. 1–6

[8] M. Havasi, R. Peharz, J.M. Hernández-Lobato, *Minimal random code learning: Getting bits back from compressed model parameters*, in *Proc. Int. Conf. Learn. Representations (ICLR)* (2019)

[9] G. Flamich, M. Havasi, J.M. Hernández-Lobato, Compressing images by encoding their latent representations with relative entropy coding. Adv. Neural Inf. Process. Sys. (NeurIPS) (2020)

[10] B. Isik, et al., *Adaptive compression in federated learning via side information*, in *Proc. Int. Conf. Artificial Intelligence Statistics (AISTATS)* (2024)

[11] M. Hegazy, R. Leluc, C.T. Li, A. Dieuleveut, *Compression with exact error distribution for federated learning*, in *Proc. Int. Conf. Artificial Intelligence Statistics (AISTATS)* (2024)

[12] A. Shah, W.N. Chen, J. Balle, P. Kairouz, L. Theis, *Optimal compression of locally differentially private mechanisms*, in *Proc. Int. Conf. Artificial Intelligence Statistics (AISTATS)* (2022)

[13] P. Cuff, Distributed channel synthesis. IEEE Trans. Inf. Theory (T-IT) (2013)

[14] D. Bergström, O. Günlü, *Deep randomized distributed function computation (DeepRDFC): Neural distributed channel simulation*, in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (Michigan, 2025)

[15] K. Sayood, *Introduction to Data Compression*, 4th edn. (Morgan Kaufmann, 2017)

[16] X. Niu, B. Bai, N. Guo, W. Zhang, W. Han, Rate–distortion–perception trade-off in information theory, generative models, and intelligent communications. Entropy **27** (2025)

[17] G. Zhang, J. Qian, J. Chen, A. Khisti, Universal rate-distortion-perception representations for lossy compression. Adv. Neural Inf. Process. Sys. (NeurIPS) **34**, 11517–11529 (2021)

[18] R. Matsumoto, Introducing the perception-distortion tradeoff into the rate-distortion theory of general information sources. IEICE Commun. Express (ComEX) **7**(11), 427–431 (2018)

[19] Y. Blau, T. Michaeli, *Rethinking lossy compression: The rate-distortion-perception tradeoff*, in *Proc. Int. Conf. Mach. Learn. (ICML)* (2019), pp. 675–685

[20] J. Chen, et al., On the rate-distortion-perception function. IEEE J. Sel. Areas Inf. Theory (JSAIT) **3**(4), 664–673 (2022)

[21] L. Theis, A.B. Wagner, *A coding theorem for the rate-distortion-perception function*, in *Neural Compression: From Information Theory to Applications–Workshop@ ICLR* (2021)

[22] M. Arjovsky, S. Chintala, L. Bottou, *Wasserstein generative adversarial networks*, in *Proc. Int. Conf. Mach. Learning (ICML)* (2017), pp. 214–223

[23] I. Goodfellow, et al., Generative adversarial nets. Adv. Neural Inf. Process. Sys. (NeurIPS) (2014)

[24] I. Gulrajani, et al., Improved training of Wasserstein GANs. Adv. Neural Inf. Process. Sys. (NeurIPS) **30** (2017)

[25] Y. Hamdi, A.B. Wagner, D. Gündüz, The rate-distortion-perception trade-off: The role of private randomness. arXiv preprint arXiv:2404.01111 (2024)

[26] A.B. Wagner, The rate-distortion-perception tradeoff: The role of common randomness. arXiv preprint arXiv:2202.04147 (2022)

[27] O. Günlü, M. Bloch, R.F. Schaefer, *Secure multi-function computation with private remote sources*, in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (Melbourne, Australia, 2021), pp. 1403–1408

[28] M. Bloch, et al., An overview of information-theoretic security and privacy: Metrics, limits and applications. IEEE J. Sel. Areas Inf. Theory (JSAIT) **2**(1), 5–22 (2021)

[29] Post-Quantum Cryptography. [Online]. Available: https://csrc.nist.gov/projects/post-quantum-cryptography. Accessed: Dec. 21, 2024

[30] A. Kalkhoran, et al., *Secure Deep-JSCC against multiple eavesdroppers*, in *Proc. IEEE Global Commun. Conf. (GLOBECOM)* (2023), pp. 3433–3438

[31] O. Günlü, M. Skorski, H.V. Poor, *Low-latency rate-distortion-perception trade-Off: A randomized distributed function computation application*, in *Joint European Conf. Networks Commun. & 6G Summit (EuCNC/6G Summit)* (Poznan, Poland, 2025)

[32] O. Günlü, R.F. Schaefer, *Low-Complexity and Reliable Transforms for Physical Unclonable Functions*, in *Proc. IEEE Int. Conf. Acoustics, Speech Signal Process. (ICASSP)* (Barcelona, Spain, 2020), pp. 2807–2811

[33] A.D. Wyner, The wire-tap channel. Bell Labs Tech. J. **54**(8), 1355–1387 (1975)

[34] Y. Polyanskiy, H.V. Poor, S. Verdú, Channel coding rate in the finite blocklength regime. IEEE Trans. Inf. Theory (T-IT) **56**(5), 2307–2359 (2010)

[35] O. Günlü, M. Bloch, R.F. Schaefer, A. Yener, *Nonasymptotic performance limits of low-latency secure integrated sensing and communication systems*, in *Proc. IEEE Int. Conf. Acoustics, Speech Signal Process. (ICASSP)* (Seoul, Republic of Korea, 2024), pp. 12971–12975

[36] G. Cervia, T.J. Oechtering, M. Skoglund, *(ϵ, n)fixed-length strong coordination capacity*, in *Proc. IEEE Inf. Theory Workshop (ITW)* (Kanazawa, Japan, 2021), pp. 1–6

[37] N. Saldi, T. Linder, S. Yüksel, Output constrained lossy source coding with limited common randomness. IEEE Trans. Inf. Theory (T-IT) **61**(9), 4984–4998 (2015)

[38] Y. Hamdi, A.B. Wagner, D. Gündüz, Rate-distortion-perception trade-off with strong realism constraints: Role of side information and common randomness. arXiv preprint arXiv:2507.14825 (2025)

[39] G. Åhlgren, O. Günlü, *Secure rate-distortion-perception trade-off over channels: A randomized distributed function computation (RDFC) application*, in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (Michigan, 2025)

[40] V. Kostina, S. Verdú, Fixed-length lossy compression in the finite blocklength regime. IEEE Trans. Inf. Theory **58**(6), 3309–3338 (2012)

[41] M.H. Yassaee, M.R. Aref, A. Gohari, *Non-asymptotic output statistics of Random Binning and its applications*, in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (Istanbul, Turkey, 2013), pp. 1849–1853

[42] M.H. Yassaee, M.R. Aref, A. Gohari, *A technique for deriving one-shot achievability results in network information theory*, in *Proc. IEEE Int. Symp. Inf. Theory (ISIT)* (Istanbul, Turkey, 2013), pp. 1287–1291

[43] V.Y.F. Tan, *Achievable second-order coding rates for the wiretap channel*, in *Proc. IEEE Int. Conf. Commun. Syst. (ICCS)* (2012), pp. 65–69

[44] J.M. Renes, R. Renner, Noisy channel coding via privacy amplification and information reconciliation. IEEE Trans. Inf. Theory **57**(11), 7377–7385 (2011)

[45] M.H. Yassaee, M.R. Aref, A. Gohari, Achievability proof via output statistics of random binning. IEEE Trans. Inf. Theory (T-IT) **60**(11), 6760–6786 (2014)

[46] T. Welling, O. Günlü, A. Yener, *Low-latency secure integrated sensing and communication with transmitter actions*, in *Proc. IEEE Int. Workshop Sign. Process. Adv. Wireless Commun. (SPAWC)* (Lucca, Italy, 2024), pp. 351–355

[47] A.E. Gamal, Y.H. Kim, *Network Information Theory* (Cambridge, U.K.: Cambridge University Press, 2011)

[48] G. Kramer, *Multi-User Information Theory* (Munich, Germany: Techn. Univ. Munich, 2018)

[49] S. Verdú, T.S. Han, A general formula for channel capacity. IEEE Trans. Inf. Theory (T-IT) **40**(4), 1147–1157 (1994)